

Statistical-Mechanical Treatment of Violations of the Double Helix in Supercoiled DNA

V. V. ANSHELEVICH, A. V. VOLOGODSKII, A. V. LUKASHIN, and
M. D. FRANK-KAMENETSKII, *Institute of Molecular Genetics,
Academy of Sciences of USSR, Moscow 123182, USSR*

Synopsis

We consider the problem of making allowance for superhelicity in the statistical-mechanical calculations of fluctuational violations of the DNA double helix. A simple model is discussed, making it possible in the calculations to use an approach based on the theory of helix-coil transition in DNA. The proposed algorithms allow calculating the effect of superhelicity on the base-pair fluctuational opening for any given sequence of nucleotides. An algorithm is also proposed allowing for the hairpin and cruciform structures in the palindromic regions of a sequence, as well as the open and helical states. The theory is used to calculate the melting curve for superhelical DNA at temperatures well below the melting point of the linear or nicked forms. The maps of opening probability are calculated for SV40 and ϕ X174 DNA using their recently published complete nucleotide sequences. The data explain well the experimental results of probing the secondary structure of these DNA by single strand-specific endonucleases.

INTRODUCTION

Over the past several years interest in the closed circular state of DNA has increased sharply. This is mainly due to the fact that it is in this state the DNA molecule performs its functions in the living cell. Extensive experimental material has been accumulated indicating that the closed circular DNA has some specific physicochemical properties. These properties vanish immediately after the introduction of a single-stranded nick into closed DNA, which proves that these properties are due to some topological constraints (see Ref. 1 and references therein).

The most important feature of the naturally occurring closed circular DNAs originating from topological constraint consists in the fact that all of them are negatively supercoiled, i.e., the sense of superhelix is opposite to the sense of the double helix. So the free energy of the closed circular DNA exceeds that of the open form by the value of the energy of superhelix formation.²⁻⁵ Since the negative superhelicity may be canceled by the opening of some portion of the DNA base pairs, the energy of a base-pair transition from the closed to opened state in the superhelical form is lower than in the linear form. Thus superhelicity may markedly change the pattern of fluctuational violations of the DNA double helix.

It was shown that the probability of base-pair opening in linear DNA may be calculated on the basis of the helix-coil transition theory.^{6,7} In this

paper we use a modification of the original helix-coil transition model that makes it possible to allow for superhelicity in the calculations of the double-helix fluctuational openings. The possibility of the occurrence of hairpin and cruciform structures in palindromic regions is also taken into account.

In the present paper we use the proposed theory to calculate the melting curves and the opening probability maps. These data supplement the recent calculations⁸ of the probability of base-pair opening and of the occurrence of cruciforms as a function of the superhelix density, which were based on the same theory.

THE MODEL

All the characteristic features of closed circular DNA stem from the fact that the two complementary strands form a linkage in the topological sense. The order of this linkage is a strict topological invariant, i.e., it cannot be changed by variations in the DNA structure other than those which introduce a backbone chain scission. This topological restriction greatly complicates the statistical-mechanical treatment.

Indeed if in the case of linear DNA it is sufficient for a complete specification of the microscopic state to attribute a base pair to one of two categories, closed or open, the situation is quite different for the closed circular molecule. To specify a microstate in the latter case, we should take into account the internal degrees of freedom for bases incorporated in the open regions. Here the free energy of the microstate depends particularly on the manner in which the two complementary chains in open regions are interwound. Indeed, the change in the way the two chains are interwound in open regions will entail two effects simultaneously. First, it will lead to a variation of the energy of the helical part of the molecule due to changed superhelix density. This is a direct consequence of the topological constraints mentioned above. Second, the free energy of the open region itself also depends on the degree of interwinding, mainly due to changes in the conformational entropy. Until now it was not known how this conformational entropy of interwinding might be computed. As a result, we are unable to solve the problem in all its complexity. So we need some simplified models.

Earlier, two simplified models for a theoretical treatment of the helix-coil equilibrium in closed circular DNA have been proposed.^{9,10} The first one⁹ was specifically proposed to treat the closed circular DNA within the helix-coil transition region. According to this model, the hydrogen bonds and stacking interactions between bases are disrupted in the open region of the closed circular form just as in the linear one. But the two complementary strands in the open region of closed circular DNA remain interwound in the same way as in the double helix. As a result, the conformational entropy of the open state in the closed circular form is lower than in the linear form. This explains why the melting temperature is higher for

the former. This model also naturally explains two other effects: the broadening of the melting range and the disappearance of the fine structure of the melting curve as one passes from the linear (or open circular) to the closed circular form.^{9,11} However, this model cannot be used for the purposes of the present study because it does not include superhelicity at all.

The second model, proposed by Laiken,¹⁰ assumes that the complementary strands in the open regions are not interwound; rather it assumes that the open and helical parts of DNA form the overall superhelix, whose free energy depends on the effective number of superhelical turns but not on the degree of denaturation itself. In contrast with the first, this second model by no means pretends to give a correct description of the helix-coil equilibrium inside the melting zone, where both the above assumptions fail. However, it may prove to be appropriate for a theoretical treatment of the problems which are within the scope of the present study, i.e., in the case when a major portion of a DNA molecule remains helical. It should be emphasized that the above assumptions which form the basis of the second model cannot be literally valid. Indeed, the greater the negative interwinding of complementary strands in opened regions, the less the energy of superhelix formation. On the other hand, such interwinding decreases the conformational entropy. It is very difficult to estimate these tendencies, but one can hope that they largely cancel each other.

So we assume that in transiently opened regions (which are formed in closed circular DNA due to thermal fluctuations), two strands are, on the average, not interwound.

As is discussed at length in Ref. 8, the above assumptions, supplemented by the experimental data on energy of superhelix formation,²⁻⁵ lead to the following expression for the free energy of superhelix formation as a function of two variables: the titratable superhelix density σ and the fraction of open base pairs ϑ ($\vartheta = m/N$, where m is the number of open pairs and N is the total number of base pairs):

$$G(\sigma, \vartheta) = 10RTN(\sigma + \vartheta)^2 \quad (1)$$

where R is the gas constant and T is the absolute temperature.

So, for the most part, our model coincides with the model originally proposed by Laiken,¹⁰ at least as long as the character of the function $G(\sigma, \vartheta)$ is not specified. But our treatment of the problem is essentially different in that it allows for DNA heterogeneity with respect to the distribution of A-T and G-C pairs, whereas Laiken¹⁰ treated the case of the homopolymer.

THE ALGORITHMS

The partition function of closed circular DNA is derived within the framework of the accepted model from the partition function of the linear molecule by multiplying each of its terms by the following factors:

$$\Phi(\vartheta) = \exp[(G(\sigma, 0) - G(\sigma, \vartheta))/RT]$$

If $G(\sigma, \vartheta)$ were a linear function of ϑ , then an incorporation of such factors would just lead to a renormalization of the equilibrium constant s . Otherwise, specific algorithms should be elaborated which make it possible to calculate the partition function of heterogeneous superhelical DNA.

Algorithm 1

We will restrict ourselves to a treatment of the simplest heteropolymer model, without any allowance for the loop formation in open regions. Let the chain consist of $N + 1$ base pairs. Denote by $g_n(m)$ and $f_n(m)$ the conditional partition functions for the first n base pairs, provided that the base pair with the order number $n + 1$ is open and closed correspondingly and among the base pairs from $n + 2$ to $N + 1$ there are exactly m open base pairs. For these quantities, the following recursion relations are valid:

$$\begin{aligned} g_n(m) &= g_{n-1}(m+1) + \hat{\sigma} s_n f_{n-1}(m+1) \\ f_n(m) &= g_{n-1}(m) + s_n f_{n-1}(m) \\ 1 \leq n \leq N, \quad 0 \leq m \leq N-n \end{aligned} \quad (2)$$

where $g_0(m) = \Phi(m/N)$ and $f_0(m) = \Phi[(m-1)/N]$, ($0 \leq m \leq N$), $\hat{\sigma}$ is the cooperativity factor, and s_n is the equilibrium constant for the n th base pair.

The quantity $g_N(0)$ presents the ordinary partition function for the chain consisting of N base pairs. Therefore, recursion equations (2) solve the problem in principle. However, practically they cannot be used for the case of large N values because of the appearance of very large numbers. This difficulty may be overcome in the following way (see Ref. 12).

Rewrite Eq. (2) as follows:

$$\begin{aligned} g_n(m) &= g_{n-1}(m+1) + \hat{\sigma} s_n f_{n-1}(m+1) \\ f_{n-1}(m+1) &= g_{n-2}(m+1) + s_{n-1} f_{n-2}(m+1) \\ g_{n-1}(m) &= g_{n-2}(m+1) + \hat{\sigma} s_{n-1} f_{n-2}(m+1) \end{aligned}$$

where $n \geq 2$.

Excluding $f_{n-1}(m+1)$ and $f_n(m+1)$, one can obtain

$$g_n(m) = s_n g_{n-1}(m) + g_{n-1}(m+1) - s_n(1 - \hat{\sigma}) g_{n-2}(m+1) \quad (3)$$

Define the values

$$G_n(m) = \frac{g_n(m)}{g_{n-1}(m)}, \quad F_n(m) = \frac{g_n(m)}{g_{n-1}(m+1)}$$

Dividing Eq. (3) by $g_{n-1}(m)$ and $g_{n-1}(m+1)$, one can obtain the following recursion equations for the quantities $G_n(m)$ and $F_n(m)$:

$$\begin{aligned} G_n(m) &= s_n + \frac{G_{n-1}(m+1)}{F_{n-1}(m)} - s_n(1 - \hat{\sigma}) \frac{1}{F_{n-1}(m)} \\ F_n(m) &= s_n \frac{F_{n-1}(m)}{G_{n-1}(m+1)} + 1 - s_n(1 - \hat{\sigma}) \frac{1}{G_{n-1}(m+1)} \end{aligned} \quad (4)$$

The partition function $g_N(0)$ may be calculated from the equation:

$$\ln g_N(0) = \sum_{n=0}^N \ln G_n(0) \quad (5)$$

The degree of helicity Θ is

$$\Theta = \frac{1}{N} \frac{\partial \ln g_N(0)}{\partial \ln s} = \frac{1}{N} \sum_{n=0}^N \frac{\partial \ln G_n(0)}{\partial \ln s}$$

where $\partial/\partial \ln s$ denotes $\sum_{i=1}^N \partial/\partial \ln s_i$.

Let us introduce the quantities

$$Q_n(m) = \frac{\partial \ln G_n(m)}{\partial \ln s}$$

and

$$P_n(m) = \frac{\partial \ln F_n(m)}{\partial \ln s}$$

From Eq. (4) one can obtain the following recursion relations for $Q_n(m)$ and $P_n(m)$:

$$Q_n(m) = [G_n(m)F_{n-1}(m)]^{-1} [s_n F_{n-1}(m) - s_n(1 - \hat{\sigma}) + G_{n-1}(m+1)Q_{n-1}(m+1) - (G_{n-1}(m+1) - s_n(1 - \hat{\sigma}))P_{n-1}(m)] \quad (6)$$

$$P_n(m) = [F_n(m)G_{n-1}(m+1)]^{-1} [s_n F_{n-1}(m) - s_n(1 - \hat{\sigma}) - (s_n F_{n-1}(m) - s_n(1 - \hat{\sigma}))Q_{n-1}(m+1) + s_n F_{n-1}(m)P_{n-1}(m)]$$

These recursion relations and Eq. (4) make it possible to calculate Θ without calculating the partition function. It follows from Eq. (5) that

$$\Theta = \frac{1}{N} \sum_{n=0}^N Q_n(0)$$

The proposed algorithm is absolutely rigorous but time-consuming. It increases the computer time in proportion to N^2 . So we have elaborated the second algorithm, which is not so rigorous, but makes it possible to calculate the equilibrium characteristics of DNA with a computer time which is proportional to N .

Algorithm 2

This algorithm is based on the natural assumption that the terms corresponding to the equilibrium quantity of open base pairs $\langle m \rangle$, or to some near values, make the main contribution to the partition function. In accordance with this assumption, superhelicity should change the free energy of base-pair opening by the value [see Eq. (1)]

$$\delta G = 20RT(\sigma + \langle m \rangle/N) \quad (7)$$

This leads to just a renormalization of equilibrium constants s_i . For such

renormalized s_i values, the value $\langle m \rangle$ may be calculated by standard helix-coil transition methods, and using this $\langle m \rangle$ value the new value of δG may be calculated by Eq. (7), and so on. So one may hope to obtain self-consistent values of the quantities $\langle m \rangle$, and δG , and of all the equilibrium characteristics of supercoiled DNA. However, the procedure of successive approximations has proved to be inapplicable because of its divergence. So we have searched for the self-consistent values in the following way.

First of all, some interval $[a, b]$ of m values was chosen which included the desired $\langle m \rangle$ value (e.g., the interval $[0, N\sigma]$). Then using Eq. (7), the δG value for $\langle m \rangle = (b - a)/2$ was calculated. For this δG value a new value of $\langle m \rangle$ was calculated using the methods of helix-coil transition theory. If this $\langle m \rangle$ value proved to be less than $(b - a)/2$, we concluded that the desired $\langle m \rangle$ value lay in the left part of the original interval. In the opposite case, it was presumed to lie in the right part. Repeating this procedure we obtained self-consistent equilibrium characteristics of closed circular DNA to any present accuracy.

As expected, this algorithm gave results which practically coincided with the results obtained by rigorous algorithm 1. Specifically, the mean opening probability for SV40 DNA equaled 6.04×10^{-4} as calculated by algorithm 1 and 6.11×10^{-4} in the case of algorithm 2 for $-\sigma = 0.05$. For $-\sigma = 0.10$, the values were 0.0219 and 0.0220, respectively.

While calculating the equilibrium characteristics of DNA with a given nucleotide sequence, the closed circularity was taken into account only by introducing the energy of superhelix formation. The calculations were actually performed for a linear sequence; and to decrease the end effects, two additional base pairs were added to the ends of the chain with extremely high equilibrium constants $s = 10^6$.

Note that algorithm 2 makes it possible to consider the more adequate DNA model, the helix-coil equilibrium model with allowance for loop formation in the open regions. To calculate the various equilibrium characteristics of closed circular DNA (which have been presented in separate paper⁸ as well as in the present one), we have used the method of Fixman and Freire¹³ for given δG values within the framework of algorithm 2.

Algorithm Allowing for Hairpin States

The traditional two-state model is insufficient when it comes to the statistical-mechanical properties of supercoil DNA, even within the framework of above simplified assumptions. Especially in the case of palindromic sequences (i.e., sequences with a twofold symmetry), hairpin (cruciform) structures may be formed with the highest probability in supercoiled DNA. Indeed, the completely helical state becomes unstable due to supercoiling energy. On the other hand, although open regions decrease the energy of superhelix formation, their occurrence at physio-

logical temperatures leads to an increase in the free energy associated with the helix-coil equilibrium. The occurrence of hairpin (cruciform) structures in palindromic regions decreases superhelicity as effectively as the formation of open regions but does not lead to the formation of a large number of open base pairs. At the same time the formation of hairpins, particularly short ones, is unfavorable because of end effects. It is the end effects that make the occurrence of hairpin and cruciform structures in linear DNA completely improbable under any conditions. But the situation may be quite different in the case of supercoiled DNA.

To allow for hairpin and cruciform structures along with the closed and open states, we have elaborated a specific algorithm. As in the previous section, we have actually considered a linear molecule and used algorithm 2 to take superhelicity into account. In doing so we naturally assumed that superhelicity changes the free energy of a base-pair opening by the value δG for all closed base pairs other than those belonging to the hairpins.

Now let us consider the double-stranded heterogeneous DNA molecule containing N base pairs. We will specify every n th base pair by the number K_n , which equals 1 if the base pair belongs to an open region and 2 if it belongs to a helical region. Let the sequence contain M palindromic regions and denote the order number of the first base pair of the m th palindrome by α_m and the last one by β_m . We assume that palindromes do not overlap, i.e., that any two adjacent palindromic regions are separated by at least one base pair which belongs to neither. If this was not the case and two or more palindromic regions *did* overlap, we retained only the one with the maximum probability of occurrence of a hairpin structure. Hereafter we mean by a hairpin structure any of the three possible violated states of a palindromic region: one strand (either of the two) forms the hairpin, but the other does not, or both strands form hairpins (the cruciform structure). These three conformations may be considered as a single state with a statistical weight equal to the sum of their respective statistical weights. Denote the statistical weight of this combined hairpin state for the m th palindromic region by t_m . Let the m th palindromic region be assigned with a quantity l_m which equals 1 if the hairpin structure is formed and 0 if it is not.

In summary, the n th base pair belonging to the m th palindromic region may be found in one of the following three states: helical ($l_m = 0, K_n = 2$), open ($l_m = 0, K_n = 1$), and hairpin ($l_m = 1$). If $P_n(K_n)$ is the probability of the n th base pair being in the state K_n and q_m is the probability of formation of the m th hairpin structure, then the following obvious relations should be valid:

$$P_n(1) + P_n(2) + q_m = 1, \quad \alpha_m \leq n \leq \beta_m \quad (8)$$

if the n th base pair belongs to the m th palindromic region and

$$P_n(1) + P_n(2) = 1 \quad (9)$$

if the n th base pair belongs to none of the palindromic regions.

Let us consider the n th base pair such as $n \neq \alpha_m - 1$ ($m = 1, \dots, M$) and denote by $P_n(K_n/K_{n+1})$ the probability of the n th base pair being in the state K_n under the condition that the $(n + 1)$ th base pair is in state K_{n+1} . Note that it is the above restriction that makes this definition correct, since according to our model, for all $n \neq \alpha_m - 1$ the fact that the $(n + 1)$ th base pair does not belong to a palindromic region means that the same holds for the n th base pair. Denote by $P_{\alpha_m-1}(K_{\alpha_m-1}/l_m = 1)$ the probability of the $(\alpha_m - 1)$ th base pair being in the state K_{α_m-1} under the condition that the m th hairpin structure has been formed and by $q_m(l_m = 1/K_{\beta_m+1})$ the formation probability of m th hairpin structure under the condition that the $(\beta_m + 1)$ th base pair is in state K_{β_m+1} . It should be recalled that according to our model, the $(\alpha_m - 1)$ th and $(\beta_m + 1)$ th base pairs do not belong to palindromic regions. [For the sake of simplicity we shall hereafter omit in all equations the index m in α_m , β_m , l_m , t_m , and q_m and will write simply α , β , l , t and q .] From the definition of conditional probabilities, one obtains

$$P_n(K_n) = \sum_{K_{n+1}} P_n(K_n/K_{n+1})P_{n+1}(K_{n+1}), \quad n \neq \alpha - 1 \quad (10)$$

$$P_{\alpha-1}(K_{\alpha-1}) = \sum_{K_\alpha} P_{\alpha-1}(K_{\alpha-1}/K_\alpha)P_\alpha(K_\alpha) + P_{\alpha-1}(K_{\alpha-1}/l = 1)q \quad (11)$$

$$q = \sum_{K_{\beta+1}} q(l = 1/K_{\beta+1})P_{\beta+1}(K_{\beta+1}) \quad (12)$$

where all sums contain only two terms corresponding to $K = 1$ and $K = 2$. So if conditional probabilities are known, the desired values P_n and q may be calculated with the help of a simple recursion procedure.

Let $Z_n(K_{n+1})$ be the partition function for a n base pairs beginning from the first one at a fixed state of the $(n + 1)$ th base pair and $Z_{\alpha_m-1}(l_m = 1)$ the same value for the first $\alpha_m - 1$ base pairs under the condition of a fixed m th hairpin structure. The above conditional probabilities may be expressed through the partial partition functions in the following way:

$$P_n(K_n/K_{n+1}) = s_n^{\delta_{K_n 2}} \hat{\sigma}^{\delta_{K_n 2} \delta_{K_{n+1} 1}} Z_{n-1}(K_n)/Z_n(K_{n+1}), \quad n \neq \alpha - 1 \quad (13)$$

$$P_{\alpha-1}(K_{\alpha-1}/l = 1) = (s_{\alpha-1} \hat{\sigma})^{\delta_{K_{\alpha-1} 2}} Z_{\alpha-2}(K_{\alpha-1})/Z_{\alpha-1}(l = 1) \quad (14)$$

$$q(l = 1/K_{\beta+1}) = t Z_{\alpha-1}(l = 1)/Z_\beta(K_{\beta+1}) \quad (15)$$

where δ denotes the Kroneker delta.

For the partial partition functions, we have the following obvious recursion relations:

$$Z_n(K_{n+1}) = \sum_{K_n} s_n^{\delta_{K_n 2}} \hat{\sigma}^{\delta_{K_n 2} \delta_{K_{n+1} 1}} Z_{n-1}(K_n), \quad n \neq \beta \quad (16)$$

$$Z_\beta(K_{\beta+1}) = \sum_{K_\beta} s_\beta^{\delta_{K_\beta 2}} \hat{\sigma}^{\delta_{K_\beta 2} \delta_{K_{\beta+1} 1}} Z_{\beta-1}(K_\beta) + t Z_{\alpha-1}(l = 1) \quad (17)$$

$$Z_{\alpha-1}(l = 1) = \sum_{K_{\alpha-1}} (s_{\alpha-1} \hat{\sigma})^{\delta_{K_{\alpha-1} 2}} Z_{\alpha-2}(K_{\alpha-1}) \quad (18)$$

Thus Eqs. (13)–(18) make it possible to calculate the conditional probabilities by carrying out a recursion procedure from left to right. In doing so we will not meet the problem of large numbers because it is the ratio of the partition functions, rather than the functions themselves, that defines the desired conditional probabilities [see Eqs. (13)–(15)]. The backward processing from right to left using Eqs. (10)–(12) gives the final probability distribution.

RESULTS OF CALCULATIONS

Using the above theory we have performed the calculations of the probability of fluctuational violation of the double helix as a function of superhelix density σ . The results of these calculations are presented and thoroughly discussed in a separate paper.⁸ Here we present the results of some other calculations which were not included in Ref. 8.

The above model leads to definite predictions as to the character of melting of superhelical DNA at temperatures well below the melting temperature of linear and nicked forms. Figure 1 illustrates this point. It presents the results of calculations of the early stage of melting of SV40 DNA for two values of superhelix density. The native superhelix density of this DNA¹ ($-\sigma = 0.057$) corresponds to curve a in Fig. 1. One can see that the theory predicts a small but measurable change in the degree of helicity of superhelical DNA at temperatures well below the melting range of the linear form. As expected, the degree of denaturation is equal to $-\sigma$ in the vicinity of the melting point of the linear form. Unfortunately, so far no one has measured the curves of thermal denaturation of closed circular DNA with sufficient accuracy to detect the effect. Our data agree qualitatively with the results on alkali denaturation of superhelical DNA.^{14,20}

In Ref. 8 we have shown that our theory gives a good explanation of the data on the rate of cleavage of closed circular DNA by single strand-specific

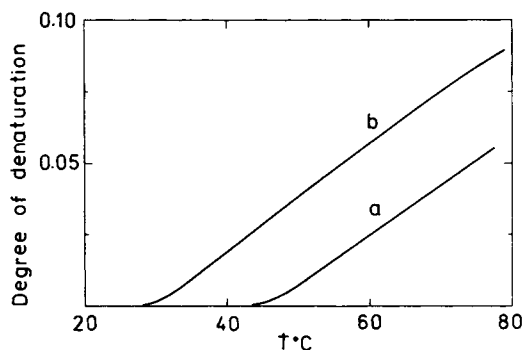


Fig. 1. Results of calculation of early melting of SV40 DNA for two values of superhelix density $-\sigma = 0.057$ (a) and $-\sigma = 0.09$ (b) at $0.1M$ Na^+ . The former σ corresponds to the native superhelix density of this DNA under standard conditions (Ref. 1).

endonucleases. In addition to the data on the overall rate of cleavage, there are reliable data on the cleavage-site specificity. It is interesting that the cleavage of SV40 DNA by S_1 endonuclease from *Aspergillus orezae* is markedly site specific,¹⁵ whereas the cleavage of ϕ x174 DNA by single strand-specific endonuclease from *Neurospora crassa* does not show any site specificity.¹⁶ This most probably reflects a difference in the patterns of fluctuational openings of the double helix for these two cases. The publication of the complete nucleotide sequences of these two DNAs^{17,18} opens the possibility of comparing the experimental results with the theoretical data on the site specificity of fluctuation violation of the double helix.

We have calculated the maps of the probability of the occurrence of open and cruciform regions in SV40 and ϕ x174 DNA for two conditions: condition I, corresponding to an ionic strength of $0.1M Na^+$, and condition II, corresponding to the DNA melting temperature decreased by $17^\circ C$. The latter corresponds to a 10-fold decrease of the ionic concentration or a considerable shift of pH from the neutral values. Condition I is close to physiological conditions and to the conditions under which the *N. crassa* endonuclease operates, whereas condition II simulates the conditions under which S_1 endonuclease operates, because this enzyme requires an acidic medium (pH 4.6). Our calculations have been performed at $37^\circ C$ and for $-\sigma = 0.057$, which corresponds to the superhelix density of both SV40 and ϕ x174 DNA under standard conditions.¹ A standard set of DNA energetic parameters for condition I was used: $U_{AT} = 8$ kcal/mol; cooperativity factor $\hat{\sigma} = 5 \times 10^{-5}$; loop-weighting factor $\alpha = 1.5$; $t_{AT} = 64.8^\circ C$; $t_{GC} = 106.2^\circ C$. For condition II the last two quantities were decreased by $17^\circ C$.

Figures 2 and 3 show the results of the calculations. Only the probability of the base-pair opening is shown, since the probability of occurrence of

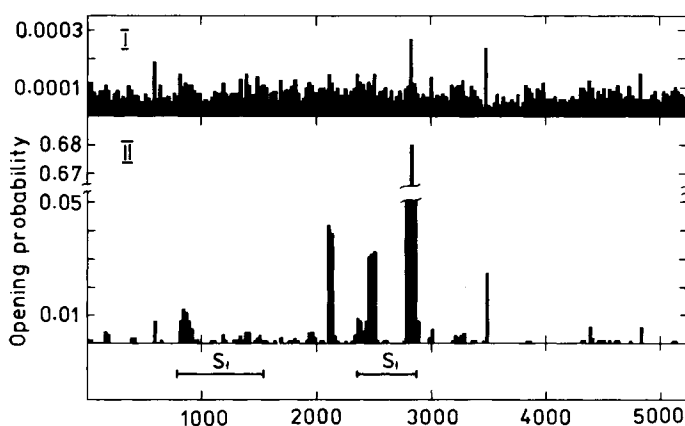
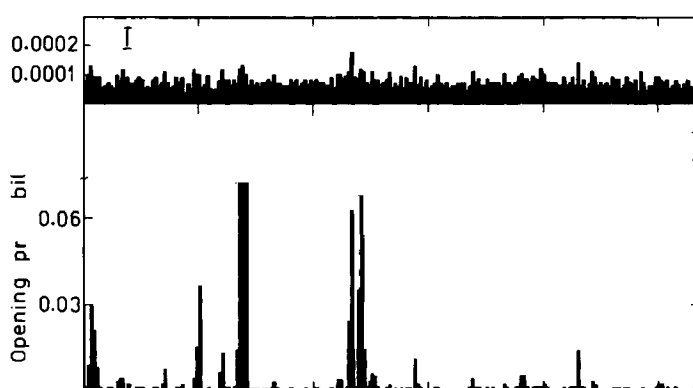


Fig. 2. Calculated opening probability maps for SV40 DNA under conditions I and II (see text). The experimental map of cleavage of this DNA by S_1 endonuclease taken from Ref. 15 is also shown.



cruciforms is negligible under the above conditions. The data demonstrate virtually the same behavior of both DNAs. Specifically, under condition I the theory predicts that both DNAs should not have any sites with a markedly increased probability of opening. The situation changes dramatically when one passes to condition II, which corresponds to an overall weakening of the double helix. Under condition II a few sites with a markedly increased probability of opening appear in both DNAs. So we see that the difference in site specificity which was experimentally revealed for the two DNAs is most probably connected with the use of different enzymes which require different environmental conditions.

The experimental map of cleavage of superhelical SV40 DNA by S_1 endonuclease obtained in Ref. 15 is shown in Fig. 2. One can see a good correlation between the sites of cleavage by this enzyme and the sites with the highest probability of base-pair opening.

CONCLUSION

In the present paper we have formulated a simple statistical-mechanical model of the influence of superhelicity on the fluctuational violations of the DNA double helix. The model is based on some assumptions which are virtually impossible to verify by direct experiment. So the adequacy of our model may be tested only by a comparison of the different predictions of the theory with the corresponding experimental results. Such a comparison is sufficiently rigorous, because our theory does not contain any adjustable parameters. In Ref. 8, as well as in the present paper, we have shown that the theory explains all the main results of probing the structure of superhelical DNA by different single strand-specific endonucleases under the simplest assumption that the rate of cleavage of a single strand of the double helix in a particular site by these enzymes is directly proportional to the probability of opening the base pairs at the site. As discussed at length in Ref. 8, these are the experiments that provide the most direct

information about the fluctuational nature of the double helix in superhelical DNA. At the same time some factors complicate the quantitative comparison of theory with experiment. The main difficulty consists in the use of very high concentrations of enzymes in these experiments. Under such conditions the effects may depend nonlinearly on the enzyme concentration (see Ref. 19), which means that the enzyme itself may perturb the DNA structure.

Thus an independent test of our theoretical predictions is highly desirable. It may consist in accurate measurements of thermal denaturation curves for superhelical DNAs and their quantitative comparison with the predicted curves presented in this paper.

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